

**SIXTH SEMESTER (CUCBCSS—UG) DEGREE (SPECIAL) EXAMINATION  
MARCH 2021****Mathematics****MAT 6B 13 (E01)—GRAPH THEORY****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 20****Maximum : 20 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

## MAT 6B 13 (E01)—GRAPH THEORY

(Multiple Choice Questions for SDE Candidates)

1. Let  $T = (V, E)$  be a tree such that  $|V| \geq 2$ . Then  $T$  has :
  - (A) At least two pendant vertices.
  - (B) At least three pendant vertices.
  - (C) At least four pendant vertices.
  - (D) No pendant vertices.
2. Let  $T$  be a tree. Then which of the following statements are true :
  - (A) There exists two paths between every pair of vertices.
  - (B)  $T$  is minimally connected.
  - (C)  $T$  has  $n$  vertices and  $n - 2$  edges.
  - (D)  $T$  has  $n$  vertices and  $n$  edges.
3. Let  $G = (V, E)$  be a connected graph and let  $v \in V$ . Then  $v$  is a central point if :
  - (A)  $e(v) > r(G)$ .
  - (B)  $e(v) < r(G)$ .
  - (C)  $e(v) = r(G)$ .
  - (D)  $e(v) \geq r(G)$ .
4. Let  $G = (V, E)$  be a graph such that  $|E| = 8$  and  $\deg(v) = 2$  for all  $v \in V$ . Then  $|V| = \underline{\hspace{2cm}}$ .
  - (A) 6.
  - (B) 2.
  - (C) 8.
  - (D) 10.
5. Let  $G_1$  be a  $(p_1, q_1)$  graph and  $G_2$  be a  $(p_2, q_2)$  graph. Then  $G_1 + G_2$  is :
  - (A)  $(p_1 + p_2, q_1 + q_2)$  graph.
  - (B)  $(p_1 q_1, p_2 q_2)$  graph.
  - (C)  $(p_1 + p_2, q_1 + q_2 + p_1 p_2)$  graph.
  - (D)  $(p_1 - p_2, q_1 - q_2)$  graph.
6. For what values of  $n$  is the graph  $K_n$  Eulerian ?
  - (A)  $n$  is a even number.
  - (B)  $n$  is an odd number.
  - (C)  $n$  is a composite number.
  - (D) None of these.
7.  $K_{m,n}$  is Eulerian if and only if :
  - (A)  $m$  and  $n$  are even.
  - (B)  $m$  and  $n$  are odd.
  - (C)  $m$  is even and  $n$  is odd.
  - (D)  $m$  is odd and  $n$  is even.

8. Let  $G_1 = (V_1, E_1)$  and  $G_2 = (V_2, E_2)$  be two graphs. Let  $f$  be an isomorphism from  $G_1$  to  $G_2$ . Let  $v \in V_1$ . Then which of the following statements are true ?
- (A)  $\deg(v) = \deg(f(v))$ . (B)  $\deg(v) > \deg(f(v))$ .  
 (C)  $\deg(v) < \deg(f(v))$ . (D)  $\deg(v) \neq \deg(f(v))$ .
9. The number of perfect matchings in the complete bipartite graph  $K_{m,n}$  is :
- (A)  $n$ . (B)  $n!$ .  
 (C)  $(n-1)!$ . (D)  $n^2$ .
10. If  $G$  is a plane  $(p, q)$  graph with  $r$  faces and  $k$  components then  $p - q + r - k = \underline{\hspace{2cm}}$ .
- (A) 1. (B) 2.  
 (C) 3. (D) 4.
11. If a  $(p_1, q_1)$  graph and  $(p_2, q_2)$  graph are homoeomorphic, then  $p_1 - p_2 + q_2 - q_1 = \underline{\hspace{2cm}}$ .
- (A) 1. (B) 0.  
 (C) 2. (D) 3.
12. If a  $(p, q)$  graph is self dual, then  $2p - q = \underline{\hspace{2cm}}$ .
- (A) 1. (B) 0.  
 (C) 2. (D) 4.
13. The origin and terminus of a longest path in a tree has degree :
- (A) 0. (B) 1.  
 (C) 2. (D) 3.
14. Every block of a tree is :
- (A)  $k_2$ . (B)  $k_3$ .  
 (C)  $k_4$ . (D)  $k_5$ .
15. Which of the following statements are true ?
- (A) Every Hamiltonian graph is 0-connected.  
 (B) Every Hamiltonian graph is 1-connected.  
 (C) Every Hamiltonian graph is 2-connected.  
 (D) Every Hamiltonian graph is 3-connected.

Turn over

**SIXTH SEMESTER (CUCBCSS—UG) DEGREE (SPECIAL) EXAMINATION  
MARCH 2021**

Mathematics

MAT 6B 13 (E01)—GRAPH THEORY

Time : Three Hours

Maximum : 80 Marks

**Section A**

*Answer all questions.*

*Each question carries 1 mark.*

1. Edges with same pair of vertices are called \_\_\_\_\_.
2. Give an example of a simple graph with 4 vertices and 5 edges.
3. Define complete bipartite graph.
4. A graph without edges is called \_\_\_\_\_.
5. Define Vertex degree.
6. Define  $k$ -regular graph.
7. Define Forest
8. Define Trail.
9. Define Hamilton cycle.
10. Define connected graph.
11. Define component of a graph.
12. Define face of a plane graph.

(12 × 1 = 12 marks)

**Section B**

*Answer at least eight questions.*

*Each question carries 3 marks.*

*All questions can be attended.*

*Overall Ceiling 24.*

13. Draw the graph  $G = (V, E)$  where  $V = \{a, b, c, d, e\}$ , and  $E = \{(a, b), (b, c), (c, c), (c, d), (b, d), (d, e), (b, e), (b, e)\}$ .
14. Define Graph isomorphism.

**Turn over**

15. Define subgraphs and spanning subgraphs.
16. Define complement of a graph. Give an example.
17. Define bipartite graph, and complete bipartite graph.
18. Let  $u$  and  $v$  be distinct vertices of a tree  $T$ . Then show that there is precisely one path from  $u$  to  $v$ .
19. State a necessary and sufficient condition for a graph to be Euler.
20. What is the smallest integer  $n$  such that the complete graph  $K_n$  has at least 500 edges.
21. Is Petersen graph Euler ? Why ?
22. State Whitney's theorem.
23. Define and illustrate Jordan curve.
24. Define critical planar graph. Give an example.

(8 × 3 = 24 marks)

### Section C

*Answer at least five questions.*

*Each question carries 6 marks.*

*All questions can be attended.*

*Overall Ceiling 30.*

25. In any graph  $G$ , show that there is an even number of odd vertices.
26. Define and illustrate with example : (1) Vertex deleted subgraph ; (2) Edge deleted subgraph and (3) underlying simple graph.
27. In Petersen graph give a trail which is not a path and a walk which is not a trail.
28. Let  $G$  be a  $k$ -regular graph, where  $k$  is an odd number. Prove that the number of edges in  $G$  is a multiple of  $k$ .

29. Draw the graph whose adjacency matrix is given by

$$\begin{bmatrix} 2 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 2 & 1 \\ 1 & 1 & 2 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

30. Define incidence matrix of a graph and illustrate it with an example.

31. A connected graph  $G$  is Euler if and only if it is a union of edge disjoint cycles.
32. Let  $u$  and  $v$  be distinct vertices of a tree  $T$ . Then show that there is precisely one path from  $u$  to  $v$ .
33. Prove that any tree with at least two vertices is a bipartite graph.

(5 × 6 = 30 marks)

### Section D

*Answer any **one** question.  
Each question carries 14 marks.*

34. Let  $T$  be a loop less graph. Then show that  $T$  is a tree if and only if for any distinct vertices  $u$  and  $v$  of  $T$ , there is precisely one path from  $u$  to  $v$ .
35. (a) Given any *two* vertices  $u$  and  $v$  of a graph  $G$ , show that every  $u - v$  walk contains a  $u - v$  path.
- (b) Define join of two graphs. Also show that  $K_{m,n} = \overline{K_m} + \overline{K_n}$ .
36. (a) Define Hamilton path and Hamiltonian cycle. Also give an example of a graph which is  $n$  not Hamiltonian but having a Hamiltonian path.
- (b) Let  $G$  be a simple graph with  $n$  vertices and let  $u$  and  $v$  be non-adjacent vertices in  $G$  such that  $d(u) + d(v) \geq n$ . Then show that  $G$  is Hamiltonian if and only if  $G + uv$  is Hamiltonian.

(1 × 14 = 14 marks)